FYUGP B.Sc. PHYSICS HONOURS

PROPERTIES OF SOLIDS

Semester V · Major Elective (Materials Science)

MODULE IV: DIELECTRIC AND MAGNETIC PROPERTIES OF SOLIDS

Model questions on topics the past papers miss

Section A: 1 mark each, answer all

Susceptibility

1 Write the relation between the magnetisation M, the susceptibility χ and the field H. **(1 mark)**

Permeability

2 Write the relative permeability μ_r in terms of the susceptibility χ. **(1 mark)**

Curie law

3 State the Curie law for a paramagnetic material and name the Curie constant. **(1 mark)**

Saturation magnetisation

4 Write the saturation magnetisation M_s of a paramagnet with N atoms per unit volume, Landé factor g and total angular momentum J. **(1 mark)**

Section B: 2 marks each, answer all

Curie–Weiss law

5 Write the Curie–Weiss law for a ferromagnet above its Curie temperature T_C, and define the Curie constant C in terms of μ_0, T_C and λ. **(2 marks)**

Néel temperature

6 Write the relation between the Néel temperature T_N and θ for an antiferromagnet, in terms of the interaction constants α and β. **(2 marks)**

Clausius–Mossotti relation

7 State the Clausius–Mossotti relation between the dielectric constant ε_r and the atomic polarisability α. **(2 marks)**

Total polarisability

8 Write the total polarisability α as the sum of its three contributions, and name them. **(2 marks)**

Section C: 4 marks each, answer all

Diamagnetism

9 Derive the classical Langevin expression $\chi_{dia} = -\dfrac{N\mu_0 Ze^2}{6m}\langle r^2\rangle$ for diamagnetism, starting from the change of orbital frequency under a field. **(4 marks)**

Paramagnetism

10 Show that for $x \ll 1$ the Langevin function is $L(x) \cong x/3$, and hence derive the Curie law $\chi_{para} = C/T$. **(4 marks)**

Origin of permanent moment and g-factor

11 Explain the origin of the permanent magnetic moment of an atom. Write the Landé g-factor in terms of J, S and L, and give its value for pure spin motion. **(4 marks)**

Antiferromagnetism and ferrimagnetism

12 Distinguish between antiferromagnetism and ferrimagnetism with a sketch of the sublattice moments, and state the sign of the exchange integral for each. **(4 marks)**

Local field

13 Derive the Lorentz local field $\mathbf{E}_{loc} = \mathbf{E} + \mathbf{P}/3\varepsilon_o$ for an atom of cubic symmetry, explaining each term. **(4 marks)**

Piezoelectricity

14 Explain piezoelectricity. Why must a piezoelectric crystal have no centre of inversion? Give one example of a crystal that is piezoelectric but not ferroelectric. **(4 marks)**

Section D: 10 marks each, answer any ONE

Weiss theory

15 Explain the Weiss molecular field theory of ferromagnetism. Derive the Curie–Weiss law above T_C and show that $T_C = \lambda N g^2 \mu_B^2 J(J+1)/3k$. **(10 marks)**

Clausius–Mossotti relation and sources

16 Derive the Clausius–Mossotti relation $\dfrac{\varepsilon_r - 1}{\varepsilon_r + 2} = \dfrac{N\alpha}{3\varepsilon_o}$ from the local field. Explain the three sources of polarisability and how their contributions change with frequency. **(10 marks)**

Ferroelectricity

17 Explain ferroelectricity in barium titanate: the cubic and tetragonal structures, the polarisation catastrophe, and the Curie–Weiss-type variation $\varepsilon_r \cong \xi/(T - T_C)$ above the transition temperature. **(10 marks)**

MODULE IV: MODEL ANSWERS

For checking your work. Each answer shows the key step and the result.

Section A. 1. $M = \chi H$ (book Eq. 8.1). **2.** $\mu_r = 1 + \chi$ (book Eq. 8.7). **3.** $\chi_{para} = C/T$ with $C = \mu_0 N\mu^2/3k$ (book Eqs. 8.33, 8.34). **4.** $M_s = NgJ\mu_B$ (book Eq. 8.49).

Section B. 5. $\chi = C/(T - T_C)$ with $C = \mu_0 T_C/\lambda$ (book Eqs. 8.64, 8.65). **6.** $T_N/\theta = (\beta - \alpha)/(\beta + \alpha)$ (book Eq. 8.82). **7.** $(\varepsilon_r - 1)/(\varepsilon_r + 2) = N\alpha/3\varepsilon_o$ (book Eq. 9.12). **8.** $\alpha = \alpha_e + \alpha_i + \alpha_d$ (electronic, ionic, dipolar; book Eq. 9.32).

Section C. 9. The orbital frequency changes by $eB/2m$ (Larmor). The induced moment is $\mu_a = -(Ze^2B/4m)\langle\rho^2\rangle$; with $\langle r^2\rangle = \frac{3}{2}\langle\rho^2\rangle$ and $M = N\mu_a$ this gives Eq. 8.20 (book §8.3.1). **10.** $L(x) = \coth x - 1/x \cong x/3$ for small x (Eq. 8.32). With $M \cong N\mu^2 B/3kT$, $\chi = \mu_0 N\mu^2/3kT = C/T$ (Eqs. 8.33, 8.34). **11.** Origin: spin and orbital motion of unpaired electrons (book §8.2). $g = 1 + \dfrac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}$ (Eq. 8.37); $g = 2$ for pure spin. **12.** Antiferromagnet: equal and opposite sublattice moments, cancelling (no net moment). Ferrimagnet: unequal opposite moments, so a net moment. Exchange integral negative in both, with AB interaction dominant (book §8.6, §8.7). **13.** The applied field $\mathbf{E}_o$ plus the polarisation field $\mathbf{E}_p = -\mathbf{P}/3\varepsilon_o$ inside a cubic site, plus the field of the other polarised atoms (book Eq. 9.3 and §9.2). **14.** Piezoelectricity: stress gives a dipole (Eq. 9.39). Needs no centre of inversion (Fig. 9.6). Quartz is piezoelectric but not ferroelectric (book §9.7).

Section D. 15. Molecular field $B_E = \lambda M$ (Eq. 8.52); $x = gJ\mu_B(B + \lambda M)/kT$ (Eq. 8.62); $\chi = C/(T - T_C)$ (Eq. 8.64), with T_C from Eq. 8.66. **16.** Eq. (9.10) and (9.12) from $P = N\alpha(E + P/3\varepsilon_o)$ (Eq. 9.9). Sources: electronic (resonance at ω_o, $\alpha_e = e^2/m(\omega_o^2 - \omega^2)$), ionic ($e^2/\beta$), dipolar ($p^2/3kT$); dipolar disappears first at high frequency (book §9.4, §9.5). **17.** Cubic phase: centres of charge coincide. Tetragonal phase: Ti and Ba move up, O moves down, giving spontaneous polarisation (book §9.6, Fig. 9.5). Polarisation catastrophe when $\sum N_j\alpha_j = 3\varepsilon_o$ (Eq. 9.34). Near T_C, $\varepsilon_r \cong \xi/(T - T_C)$ (Eq. 9.38).

Mapping to the syllabus and textbook

Syllabus item	Topic	Puri and Babbar section	Model questions
17	Types of magnetism; origin of the permanent moment	8.1, 8.2	1, 2, 11
18	Diamagnetism and paramagnetism (classical); ferromagnetism (Weiss)	8.3, 8.4, 8.5	3, 4, 5, 9, 10, 15
19	Antiferromagnetism and ferrimagnetism (qualitative)	8.6, 8.7	6, 12
20	Polarisation, susceptibility, local field	9.1, 9.2	13
21	Dielectric constant, polarisability and its sources	9.3, 9.4, 9.5	7, 8, 16
22	Ferro- and piezoelectricity (qualitative)	9.6, 9.7	14, 17

Questions are drafted for the gaps in the past papers (not from any past paper). Chapter questions and problems of the book are not used.